

ADT Studio · Test build for team review

Kids Mode

A reading companion for the youngest readers — and three different ideas about how it should work. We'd like you to pick one.

▶ eliezir.github.io/adt-kids-mode-demo

Demo book: **Hyena and Raven** — Tony Lelliott & Wiehan de Jager, from African Storybook, used under its open licence. Available in English, Portuguese (Brazil) and Spanish (Uruguay).

Nothing here is final

 **This is a test build from an unmerged branch. Bugs are expected.**

We're showing it early *because* we want the direction challenged while changing it is still cheap.

Three things are deliberately temporary — please don't read them as decisions we've made.

WHAT YOU'LL SEE	WHY IT'S THERE
A List / Chat / Shelf switch, top-left	So you can compare three menu designs on the same book. A developer control — untranslated, and it disappears once one design is chosen.
Three different buddy menus	We couldn't decide on paper, so we built all three. We want you to pick.
Buddy art that isn't final	The five buddies are a working set, not a commissioned cast.

Ask 1

Which buddy menu?

Ask 2

Are PNG buddies the right approach?

Ask 3

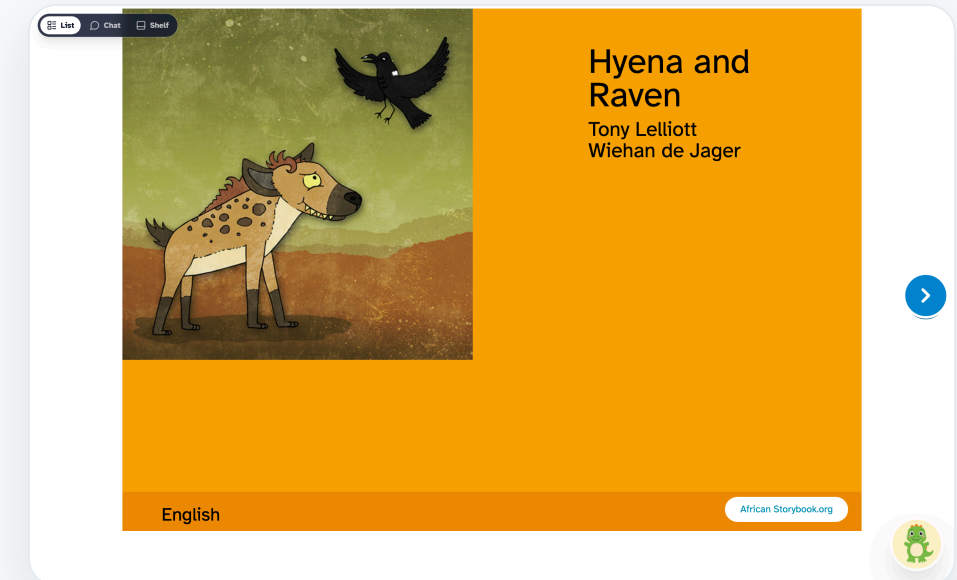
Does our plan for what's next make sense?

OVERVIEW

What Kids Mode is

A child-facing reading layer for ages 4–8 — many pre-literate or early readers, some with low vision or motor difficulties. It wraps an ordinary ADT book without altering it.

- A buddy greets the child **by name**, reads aloud, offers help, reacts to answers and celebrates the end of the book
- Designed so a child who **cannot read the labels at all** can still navigate — by icon, colour, size and position
- Every control is at least **44 × 44px**; motion respects `prefers-reduced-motion` ; nothing scrollable hides without a cue
- Ships **inside the book package** and works offline — no network, no accounts, no per-child server state



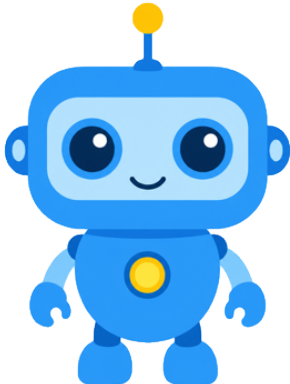
THE BUDDIES

Five characters, one chosen friend

The child picks a buddy during onboarding and can change it later. Each has its own voice preset and delivery style.



Rex
Dinosaur · onyx



Bolt
Robot · echo



Pip
Bunny · nova

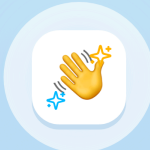


Luna
Cat · sage



Zibby
Alien · fable





Pick a reading buddy



Rex



Bolt



Pip



Luna



Zibby

This is my buddy

Kids Mode — UX feature report

Choosing is the first real decision the child makes, and it sets the tone for everything after — the voice that reads to them, the face that reacts to their answers, the character that congratulates them at the end.

The buddy starts on a **random** character each time, so no one buddy is quietly privileged by being the default.

4

Seven poses each

So the character can *react* rather than just sit there. Rex's full set, as shipped:



standing
1



signature
2



happy
3



excited
4



thinking
5



surprised
6



encouraging
7

POSE	WHEN THE CHILD SEES IT
standing	Idle, at rest in the corner
signature	The character's own hallmark pose
happy	Menu open, general warmth
excited	Correct answers, celebration, end of book
thinking	While processing or waiting
surprised	Feature reveals during onboarding
encouraging	After a wrong answer — “try again”

How they're built — plain numbered PNGs

Deliberately the simplest thing that works.

```
assets/kids-buddies/<id>/<id>_1.png ... _7.png
```

- **35 PNGs total.** No sprite sheets, no animation runtime, no 3D, no rigging — the runtime just swaps the image source
- Motion comes from **CSS**: a gentle idle bob, a pop when the expression changes. Fully disabled under reduced motion
- **Adding a sixth buddy is dropping in a folder of 7 PNGs** and one registry entry. No pipeline, no tooling, no code

That was the point — we wanted the cast to be cheap to extend *before* we knew who the cast should be.

5

buddies

7

poses each

35

PNG files, total

0

extra runtime dependencies

This is open to change

Tell us if you'd rather have something richer. The trade-off, honestly:

Today

Numbered PNGs

Trivial to add, tiny, offline-safe. But expressions are fixed — no lip-sync, no in-between motion, no reaction we didn't pre-draw.

Option

Layered / rigged (Rive, Lottie)

Real animation, lip-sync while speaking, blended emotions. Costs an animation runtime, an authoring pipeline, and a specialist per character.

Option

3D characters

The richest result, and by far the most expensive to produce, license and run on the low-end devices this programme targets.

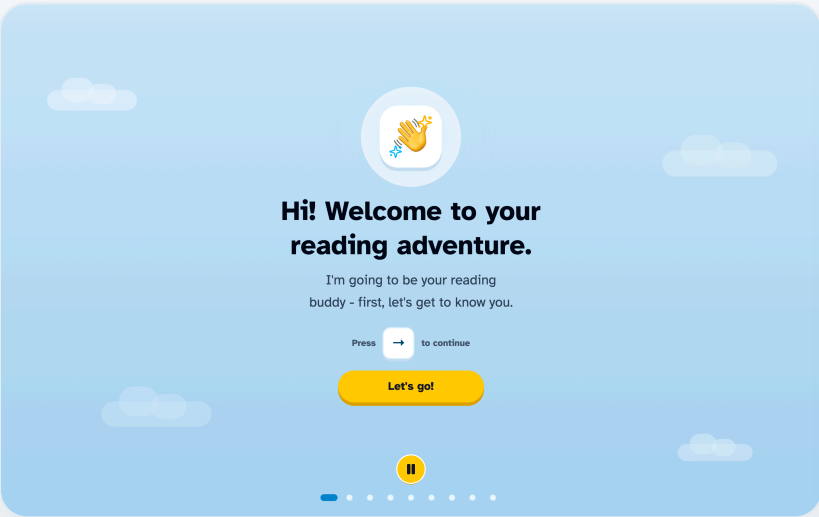
Planned either way: let people add their own buddies

A buddy library in Studio — an author uploads a name, seven PNGs and a voice preset, and that buddy appears in every book they produce. The runtime is already built for it: it accepts arbitrary image URLs, falls back safely for buddy IDs it doesn't recognise, and the “Add buddy” tile is already stubbed in the UI. The remaining work is the Studio-side library and storage. Further out, the same mechanism could let a teacher or child add a buddy — a much bigger conversation about moderation and safety.

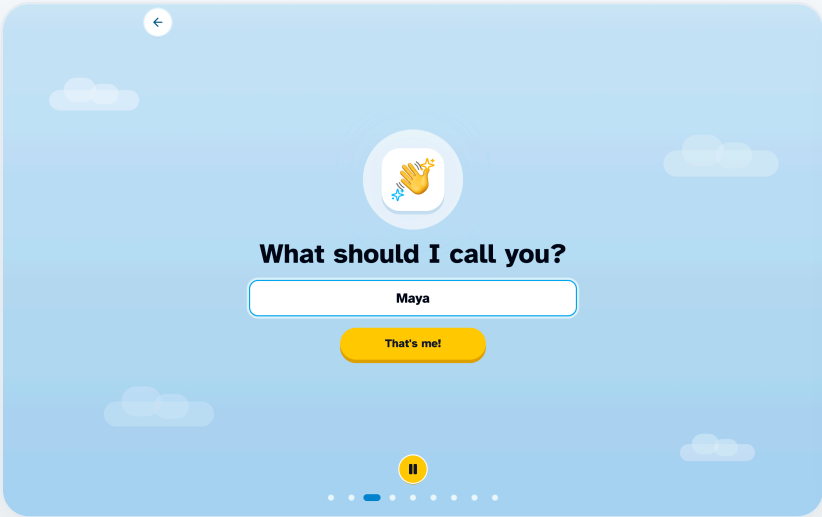
FEATURES

The buddy talks to the child

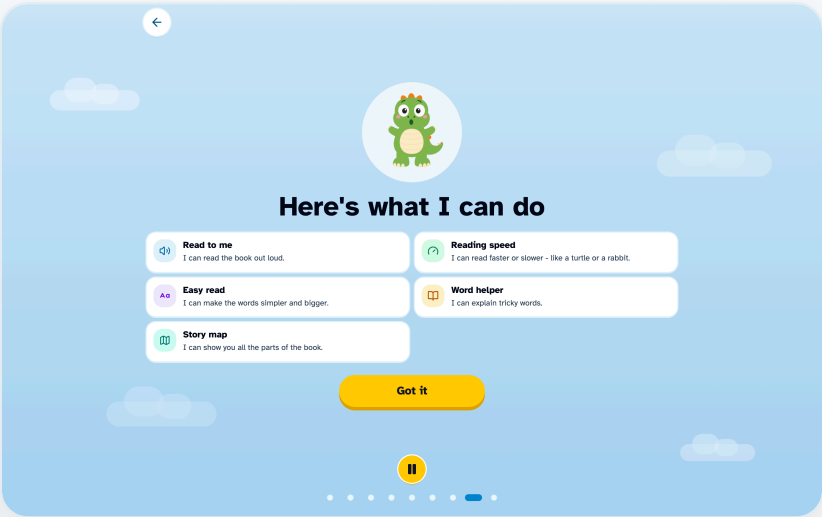
Actual audio, not just text bubbles. All speech is pre-generated into the book package and plays offline.



A narrated welcome — read aloud by default, with a pause button



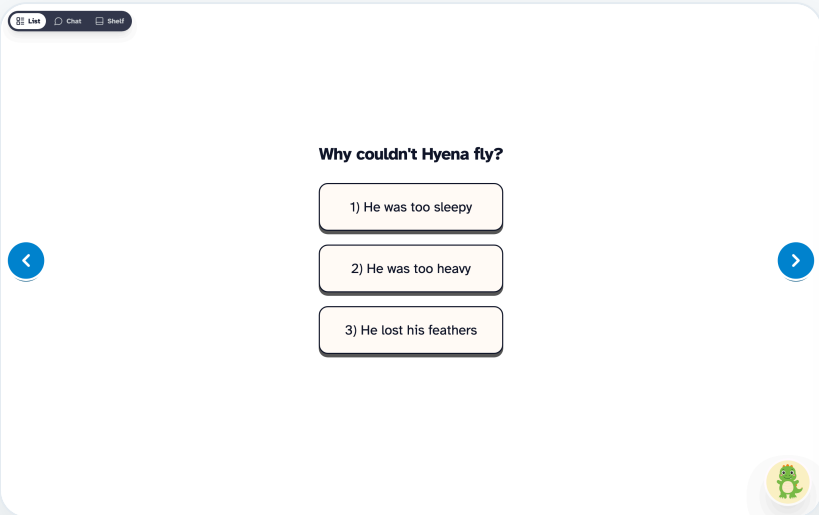
The child tells the buddy their name



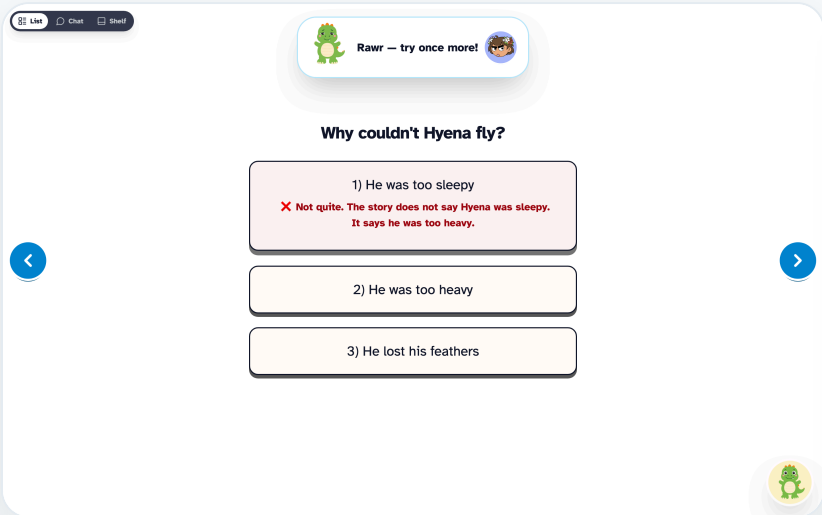
The buddy introduces what it can do

- **Greets the child by name** on arrival
- **Confirms every action out loud** — “Okay! Now I read slowly, like a turtle.”
- **Chats unprompted while reading** — randomised 45–90s gap, paused during narration, switchable off
- **Narrates the whole onboarding**, so a pre-literate child is never stranded at a screen of text

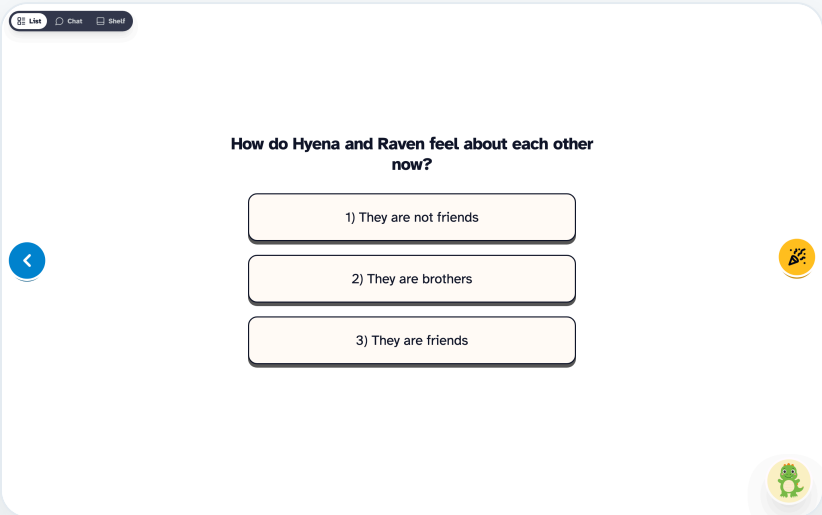
The buddy reacts to every answer



A quiz question



The buddy responds beside the child's own avatar — out loud



Finishing the book earns a celebration

Quizzes now answer on click.

Picking an option judges it immediately — verdict, explanation, sound, buddy reaction and confetti, all on one tap. The Submit button is gone for quizzes and returns as **Next** once everything is right. For a five-year-old, “choose, then find and press a second button” was a step with no meaning.

Keyboard users keep a commit step.

Arrow keys browse the options *without* judging — otherwise simply navigating would record wrong answers — and Enter commits. The buddy's reaction follows the answer just given, not the whole page.

Half the menu was invisible

The buddy's menu holds every tool in Kids Mode. The original was measurably broken.

~50%

of actions hidden below the fold —
at *every* screen size

0

cues that more existed: no scrollbar,
no fade, no arrow

416px

wide column on a 1440px display

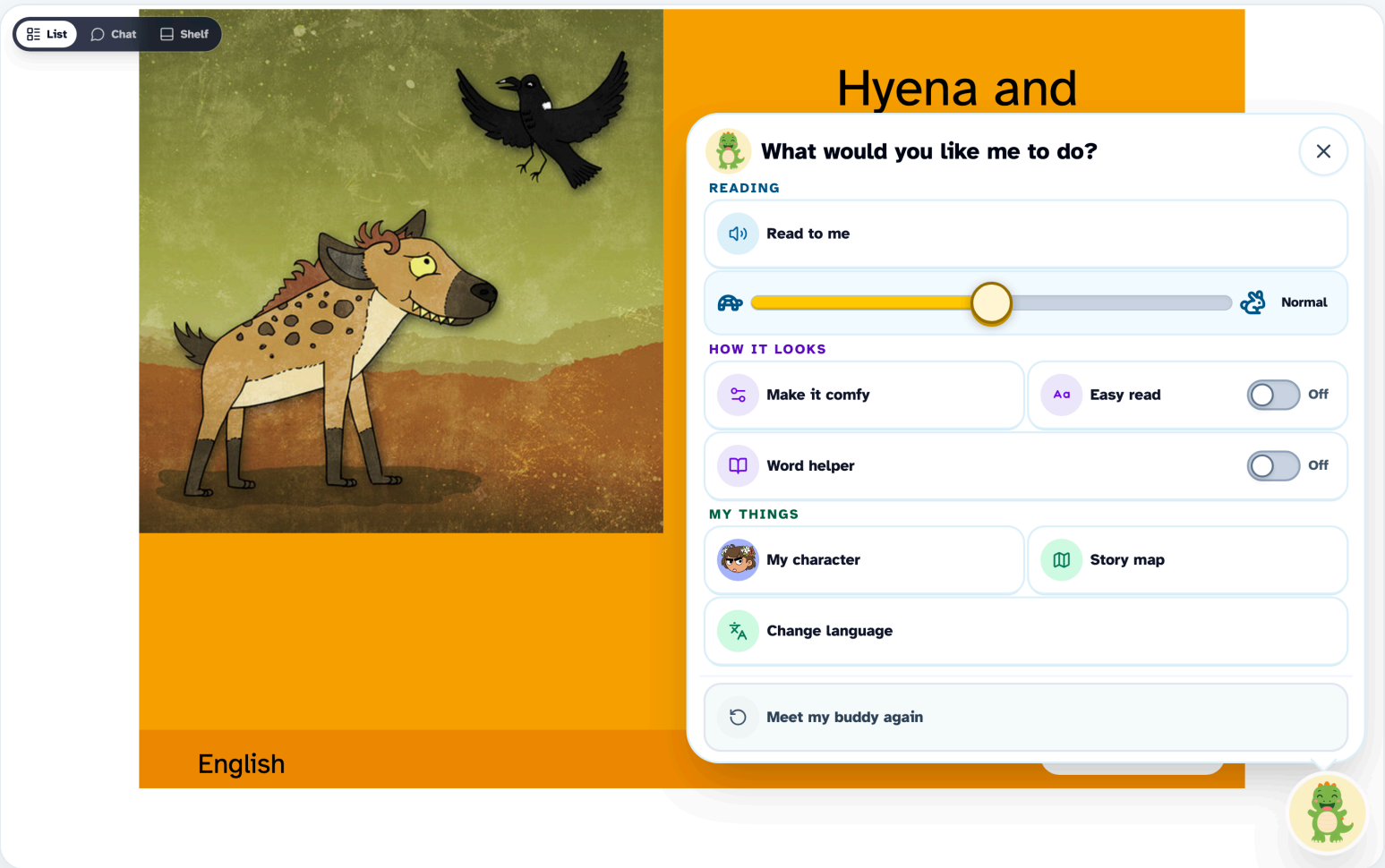
9–12

identical-looking rows in one flat list

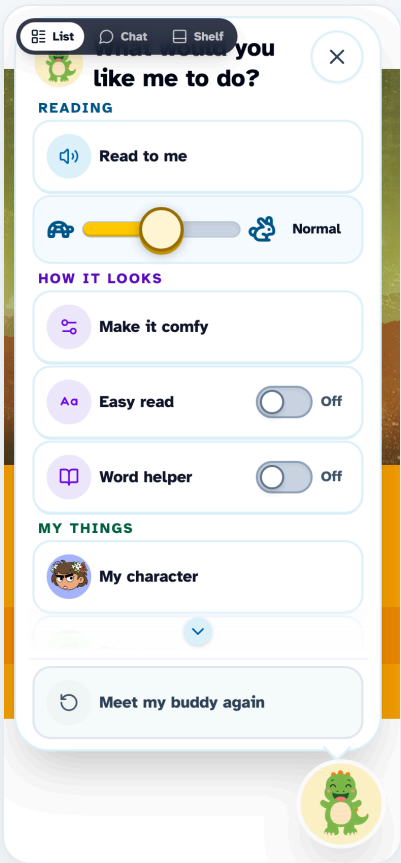
The last visible card was simply sliced in half, so a child had no way to know the rest of the menu existed. Toggle labels flipped between “Easy read on” and “Easy read off”, genuinely ambiguous about whether they described the current state or the action.

That earned a redesign — and we built three, because the right answer wasn't obvious.

List — grouped, colour-coded

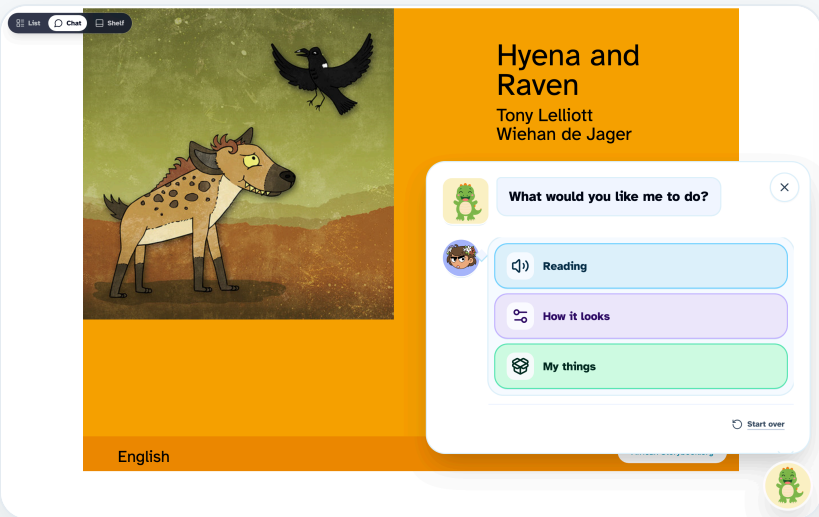


Desktop · 1440 × 900

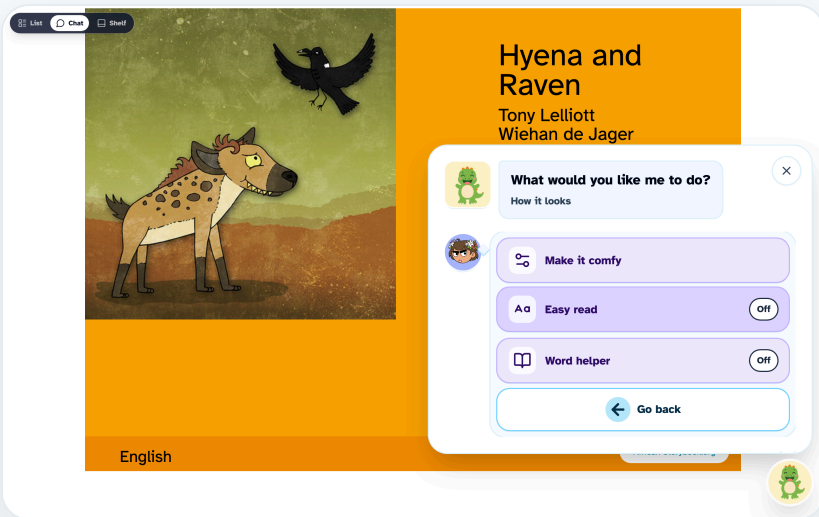


- Actions grouped under **Reading, How it looks** and **My things**, each with its own colour
- Speed is a **slider** from turtle to rabbit
- Toggles are real **switches** with a sliding knob and On/Off label
- “Meet my buddy again” — which restarts onboarding — sits apart in a quiet footer
- **No scrolling** on desktop or tablet; a fade and chevron appear where it can overflow

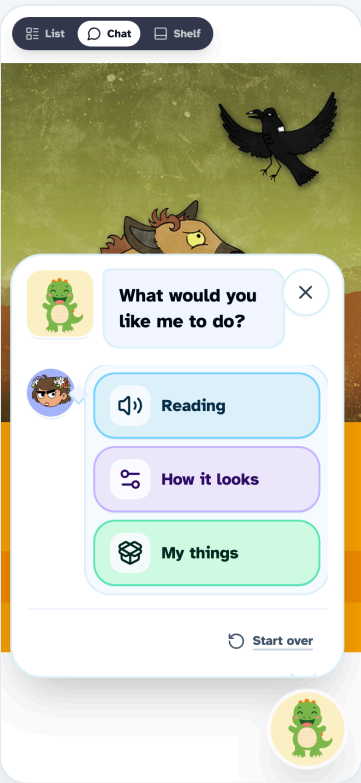
Chat — the menu as a conversation



The buddy asks — at most four big replies



Choosing a group drills in, with a big Go back

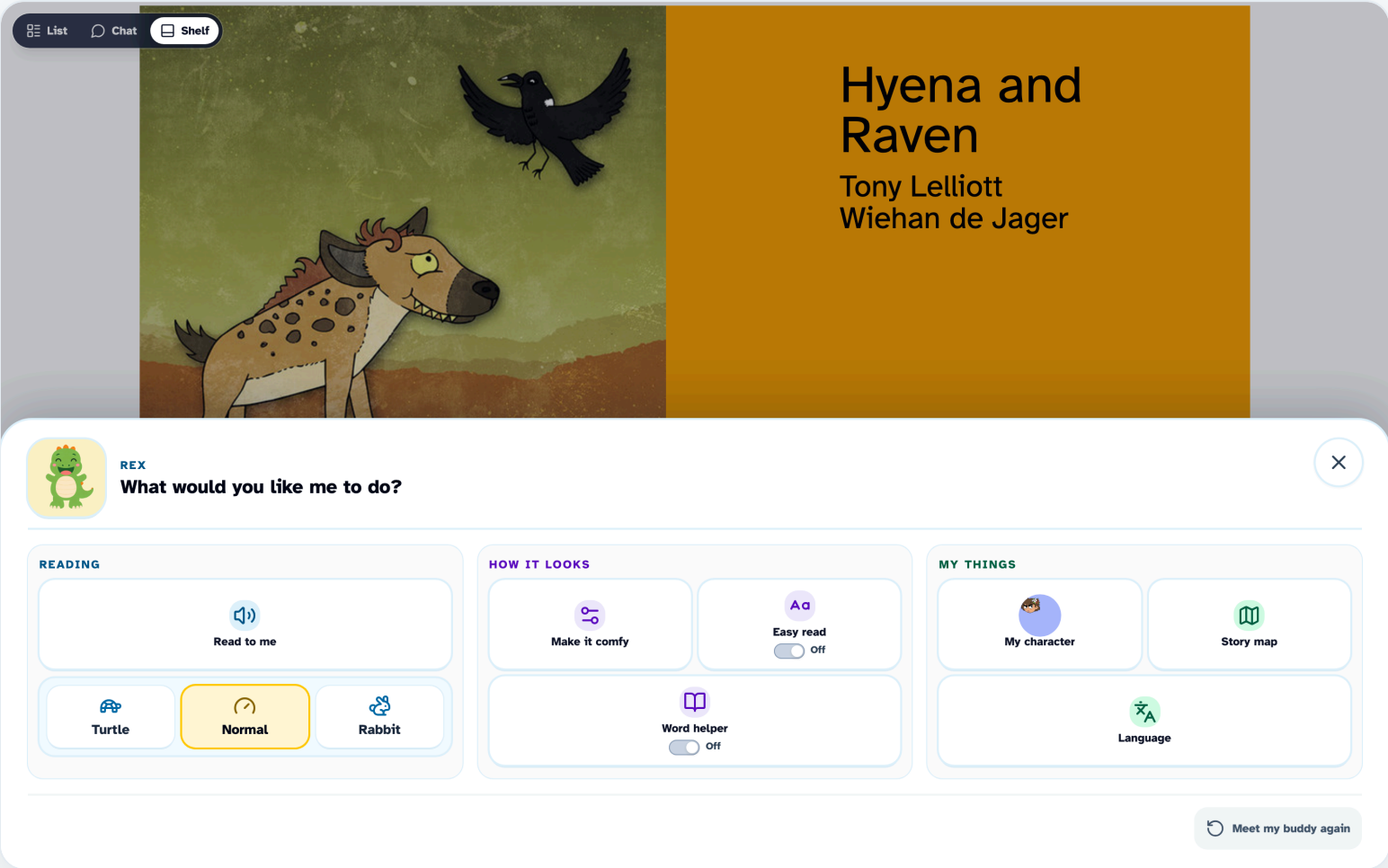


Phone

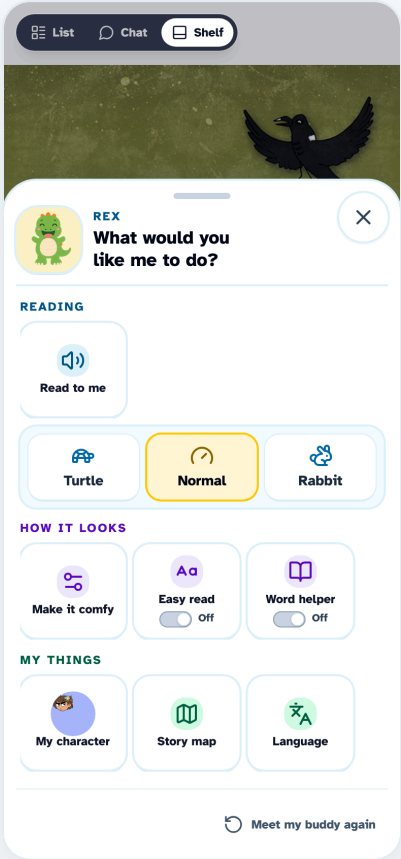
The lowest cognitive load of the three. The child answers a question instead of scanning a menu — never more than four choices on screen, ever. Transitions slide directionally and replies arrive staggered, so it reads as a conversational turn rather than a screen swap.

The cost: common actions take two taps instead of one, and it uses the least screen space of the three designs. Whether that calm is worth the extra tap is exactly what we can't judge for you.

Shelf — a full-width dock



Desktop · three group columns across the full display



- Rises from the bottom edge and **uses the whole width** the corner popover wasted
- Three group columns on desktop; a proper **bottom sheet with a grab handle** on a phone
- Its **switch-style toggles** are the clearest state indicator of any of the three designs
- Rows only gain paging arrows when they genuinely overflow

Side by side

Same book, same actions, measured on the live build.

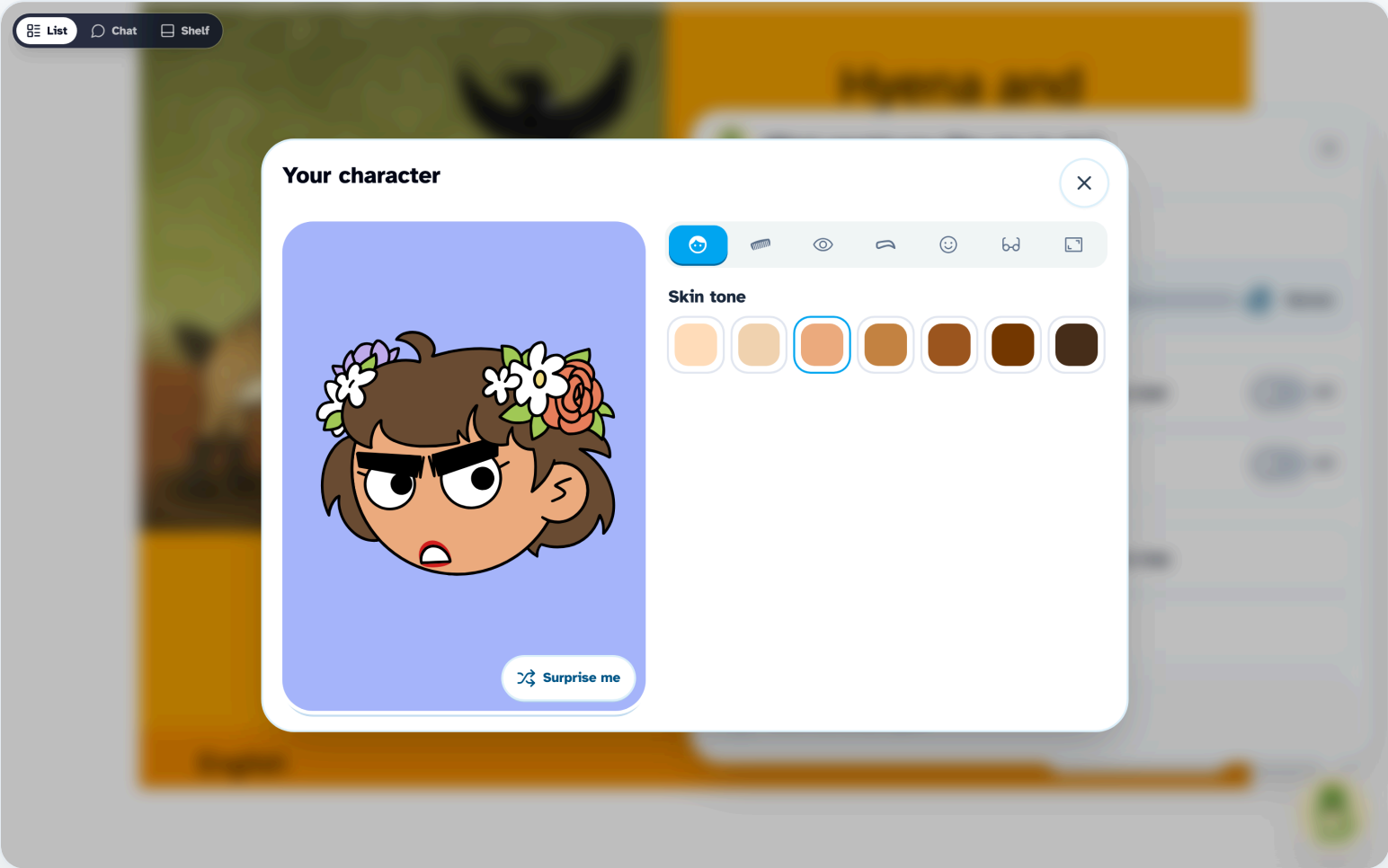
	LIST	CHAT	SHELF
Panel size at 1440 × 900	704 × 674	672 × 510	full width
Actions visible at once	9	3	12
Scrolling needed	none	none	none
Controls under 44px	0	0	0
Off-screen controls	0	0	0
Taps to a common action	1	2	1

All three clear the accessibility floor. The choice is about *feel* — which is why we need your eyes rather than our measurements.

Switch between them with the **List / Chat / Shelf** control in the top-left corner of the demo.

The child builds themselves

Separate from the buddy. This avatar then appears beside the buddy in activity reactions, on the celebration screen and in the menu.



PART	OPTIONS
Skin tone	7
Hair style	45 (26 long, 19 short)
Hair colour	14
Eyes	26
Eyebrows	15
Mouth	30
Glasses / earrings	5 / 6 + none
Features	freckles, blush, birthmark
Background	7

Rendered **offline as SVG** — no network, no image requests, no per-child asset storage. Built low-vision first: colours come first, a large live preview, big tiles, icon-only tabs, and a shuffle button for children who'd

The library, and the other styles available

We render with **DiceBear**, an open avatar library. We chose its *Adventurer* style — but the same library ships **31 styles**, all already installed. Switching is a style name and a part catalog, not a new dependency.

STYLE	CHARACTER
Adventurer in use	Illustrated, friendly, wide feature range
Avataaars	The familiar flat-vector look; very large catalog
Micah	Clean, modern, softer palette
Dylan	Bold, simple shapes, few but strong options
Big Ears	Playful and cartoonish, reads well small
Lorelei	Hand-drawn, delicate line work

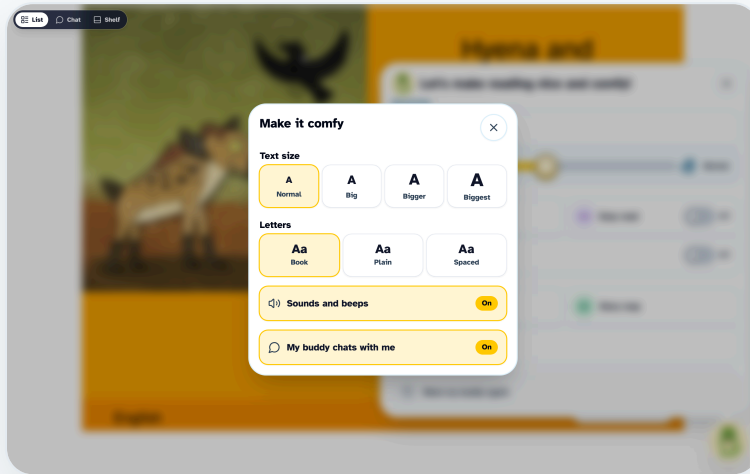
STYLE	CHARACTER
Open Peeps	Sketchy, hand-illustrated, very characterful
Notionists	Minimal line-art
Personas	Rounded, corporate-friendly
Pixel Art	Retro game sprites
Bottts	Robots — a possible route for <i>buddies</i>
Fun Emoji	Expressive faces, no body

Worth a look together. *Adventurer* was a reasonable first pick rather than a researched one, and a style like *Open Peeps* or *Big Ears* may sit better beside hand-illustrated buddies and storybook art. Two styles — **Bottts** for robots and **Fun Emoji** for faces — could also give us buddy characters from the same library.

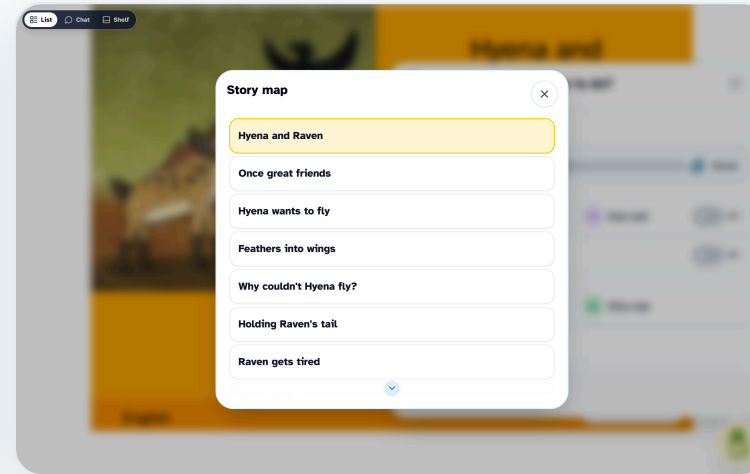
The part catalog is **data, not code**: narrowing the options, tuning skin and hair palettes per region, or swapping the style entirely are all list edits. Custom illustrated artwork could replace DiceBear behind the same builder UI.

FEATURES

Comfort, the child's way



"Make it comfy"



"Story map" — jump anywhere in the book

- **Four text sizes**, applied to the book content itself rather than just the interface
- **Three letter styles** — the book's own font, a plain sans-serif, and a spaced dyslexia-friendly treatment
- **Buddy chatter** can be switched off entirely
- **Sound effects** have their own switch — page turns and answer verdicts can be silenced without silencing the story
- The whole book **scales together**, so pictures and words stay in step

Where the voices fall short

How they work today

- Generated with the **OpenAI speech API** — one clip per line, per language
- **Baked into the book package** and cached, so regeneration is cheap and playback is fully offline
- Each buddy has a preset voice — `onyx`, `echo`, `nova`, `sage`, `fable` — plus a per-buddy style instruction shaping delivery
- Voice packs currently exist for **two languages**; the pipeline handles any language, so this is a cost question rather than a technical one

The honest problem: customisation is poor

These are **human** voices acting. We can nudge tone with instructions, but we cannot make Bolt sound like an actual robot, or Zibby like an actual alien.

A child meeting a robot that sounds like a man doing a voice is a weaker illusion than the artwork deserves.

We're still looking for a better approach. The provider layer is already abstracted — OpenAI, Azure and Gemini implementations exist — so swapping is a contained change. **ElevenLabs** looks the most promising route to genuine creature and robot timbre, including designing a voice from a description. That brings a new API dependency and cost profile, so we'd want your view before committing.

KNOWN ISSUES

What's missing, plainly

Listed so nobody files these as surprises.

Biggest gap

The buddy cannot say the child's name

The buddy *displays* “Hi Maya!” but **speaks** the generic “Hi! Tap me if you need help.” Names are unbounded and clips are pre-generated offline, so there is no clip for “Maya”.

Given the whole design rests on a personal relationship with the buddy, we think this is worth solving properly.

Not built

No voice input

The child cannot *speak* to the buddy — everything is tap or keyboard.

This is the most requested direction and the hardest to do well: it must work **offline**, which rules out cloud speech recognition and makes browser speech APIs unreliable on the devices this programme targets. We want to spike the recognition path before committing.

Also open — no dark mode yet. A calmer, low-stimulation theme is something we'd like to add for children who find the bright sky palette tiring.

Ideas we're turning over

Not a committed roadmap — this is what's on our minds, shared so you can tell us what sounds right, what sounds wrong, and what we've missed.

Close

- Interface translations.** The book text translates today, but the buddy's own words only exist in a couple of languages.
- Dark mode.** A calmer theme for children who find the bright palette tiring.
- A buddy who coaches.** After a couple of wrong answers, offer a hint rather than just encouraging another try — the explanations already exist in the book.

Solve properly

- The buddy saying the child's name.** The gap that bothers us most. Needs a real approach, not a quick patch.
- Turning pages more smoothly.** Each page is a fresh load today. Swapping pages in place would give us a page-turn animation, background ambience, and read-aloud that doesn't stop at the page edge — all from one change.
- Better voices.** Whether believable creature voices justify a new provider.

Further out

- Letting people add their own buddies** — the natural next step for personalisation, and the piece we're most curious whether you want.
- Talking to the buddy.** Genuinely hard offline, so we'd want to test the idea before promising it.
- A sign-forward experience** for deaf and hard-of-hearing children — which we don't think we should design alone.
- Small delights** — earning things, decorating your buddy, ambient sound. Firmly behind everything above.



Reading, together.

Please try the build and tell us what feels wrong for the age group, however small. Bugs are expected — treat rough edges as rough edges rather than decisions.

► eliezir.github.io/adk-kids-mode-demo

